

Act V: Different Stages

Market Structures

Module XIII: The Farmers Market

Perfect Competition



Previously in Marginal Falls

So far, we built the pieces:

- ◆ Buyers → Demand
- ◆ Sellers → Supply
- ◆ Markets → Equilibrium
- ◆ Fair Play → Surplus
- ◆ Town Crisis → Market Failure

Next: Different ways sellers can compete.

Recap.

Revenue, Costs and Profits

Total Revenue is the money the firm receives from selling output.

$$\text{TR} = P \times Q$$

Profit is what remains after paying the cost of production.

$$\text{Profit} = \text{TR} - \text{TC}$$

Revenue, Costs and Profits

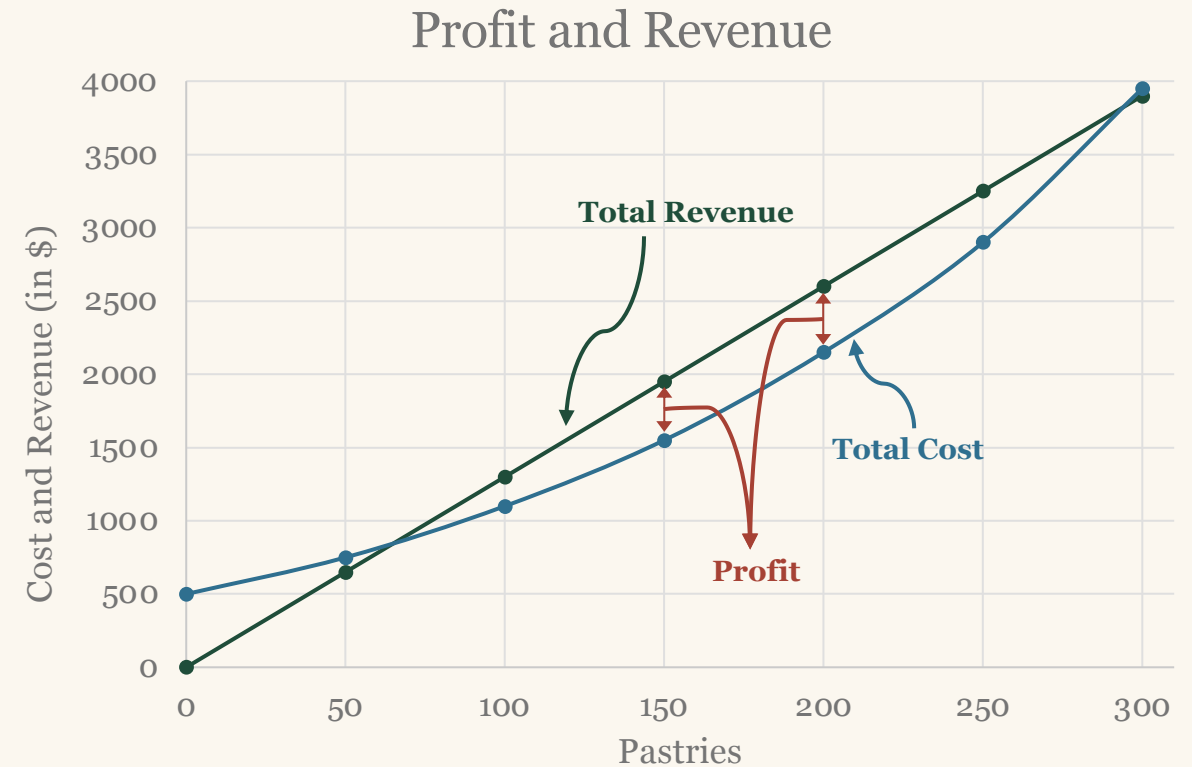
$$TR = P \times Q$$

$$\text{Profit} = TR - TC$$

Pastries (Q)	Price (P)	Total Revenue (TR)	Total Cost (TC)	Profit
0	\$13	\$0	\$500	-\$500
50	\$13	\$650	\$750	-\$100
100	\$13	\$1,300	\$1,100	\$200
150	\$13	\$1,950	\$1,550	\$400
200	\$13	\$2,600	\$2,150	\$450
250	\$13	\$3,250	\$2,900	\$350
300	\$13	\$3,900	\$3,950	-\$50

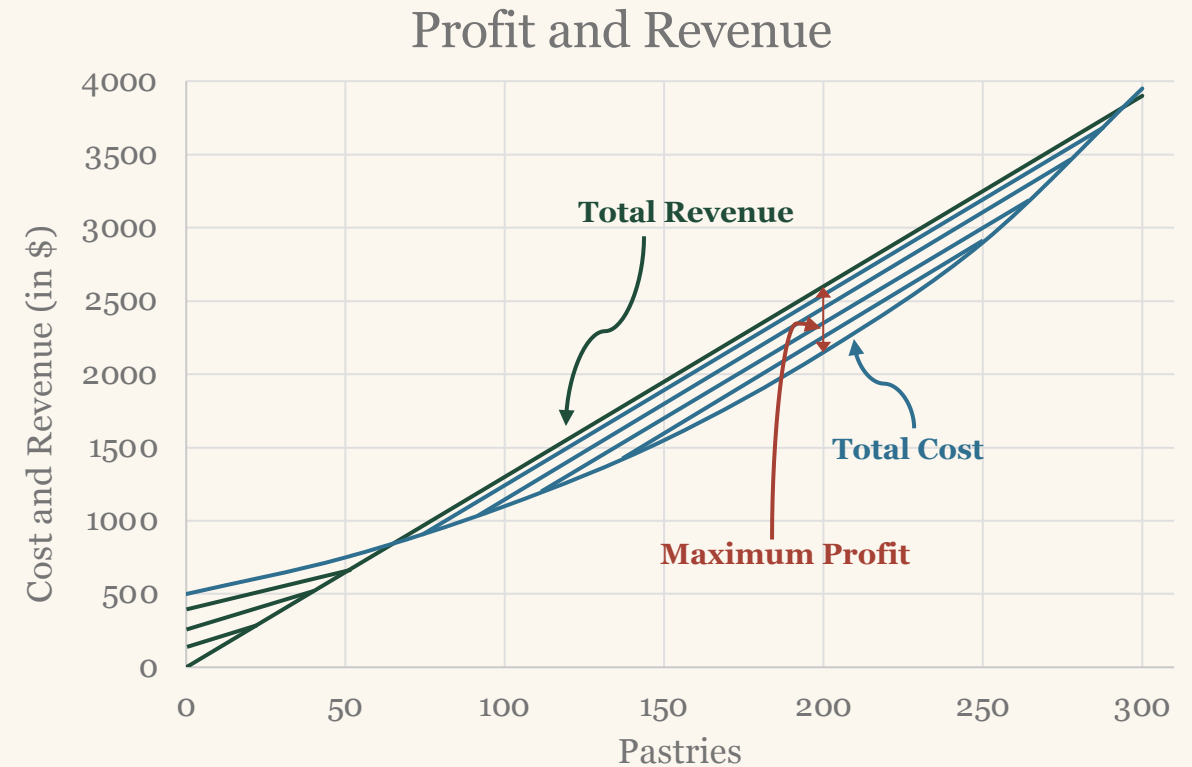
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One More Pastry

Marginal Revenue is the change in Total Revenue from selling one more unit of Pastry.

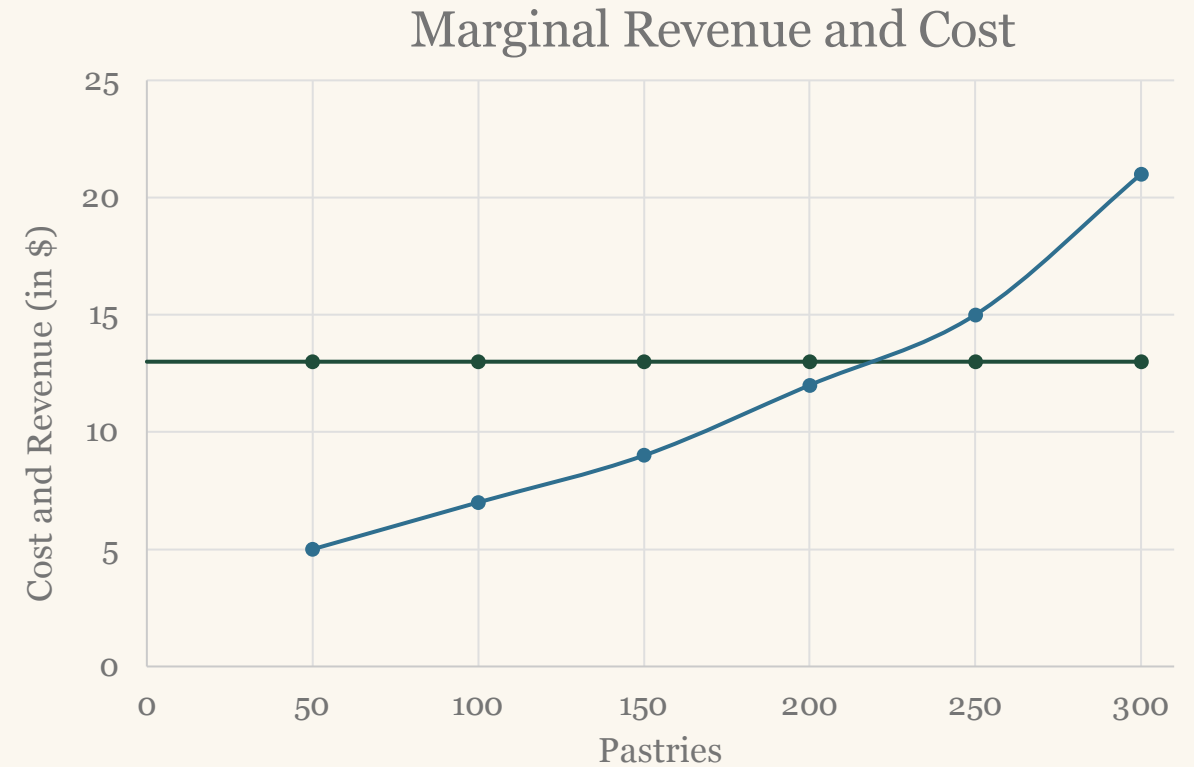
$$MR = \frac{\Delta TR}{\Delta Q}$$

Here, each extra Pastry brings in \$13, because of fixed price.

$$MR = \$13$$

Marginal Revenue

Pastries (Q)	Price (P)	Marginal Revenue (TR)	Total Cost (TC)	Marginal Cost (MC)
0	\$13	-	\$500	-
50	\$13	\$13	\$750	\$5
100	\$13	\$13	\$1,100	\$7
150	\$13	\$13	\$1,550	\$9
200	\$13	\$13	\$2,150	\$12
250	\$13	\$13	\$2,900	\$15
300	\$13	\$13	\$3,950	\$21



Optimal Production

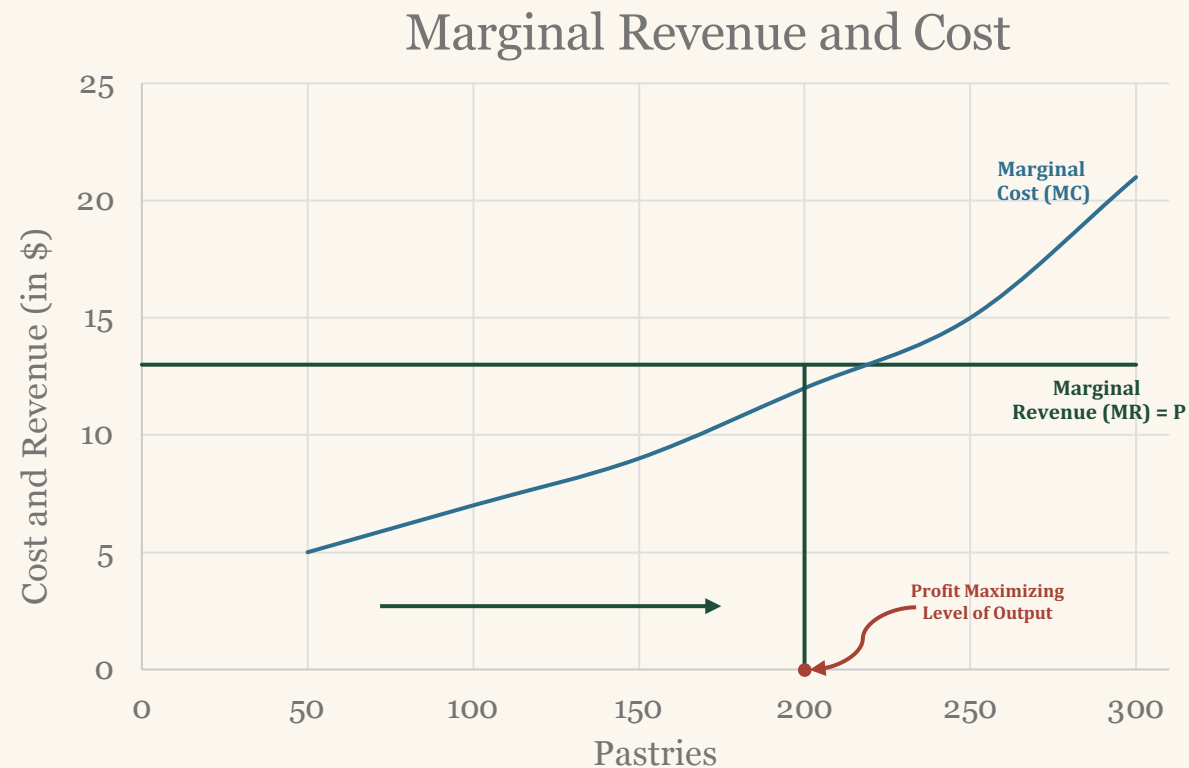
Produce as long as:

$$MR \geq MC$$

Since, $MR = P$

Produce as long as:

$$P \geq MC$$



Optimal Production

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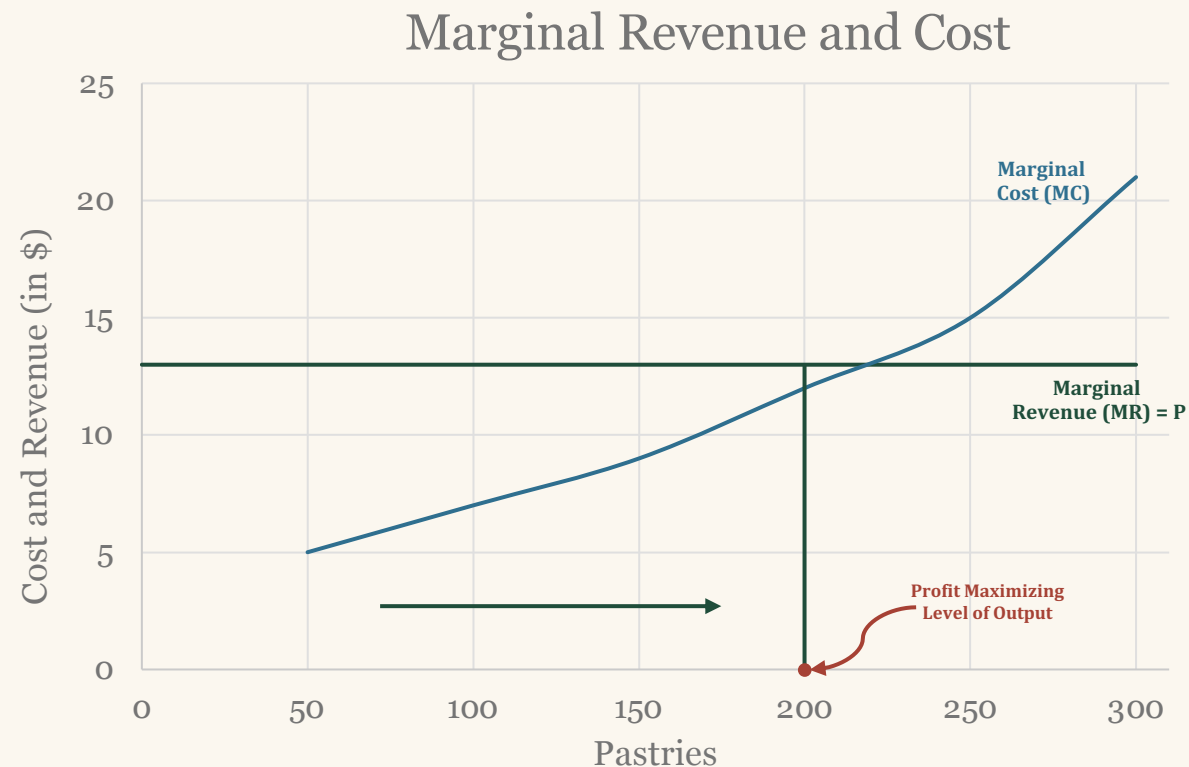
Since, $MR = P$

Produce as long as:

$$P \geq MC$$

Optimal Production,

$$Q^* = 200$$



The Mayor Leaves

The Mayor removes the price sign.

Now sellers choose prices.

But buyers can compare stalls.

If all pastries are identical, a seller who charges more loses buyers.



Can a seller really choose any price?

Set the Price

Econgames.promitkchaudhuri.com/price

Name your supermarket.

Choose a price to sell your products at.

See how the buyers react.

Market Structure

A market structure describes how sellers compete.

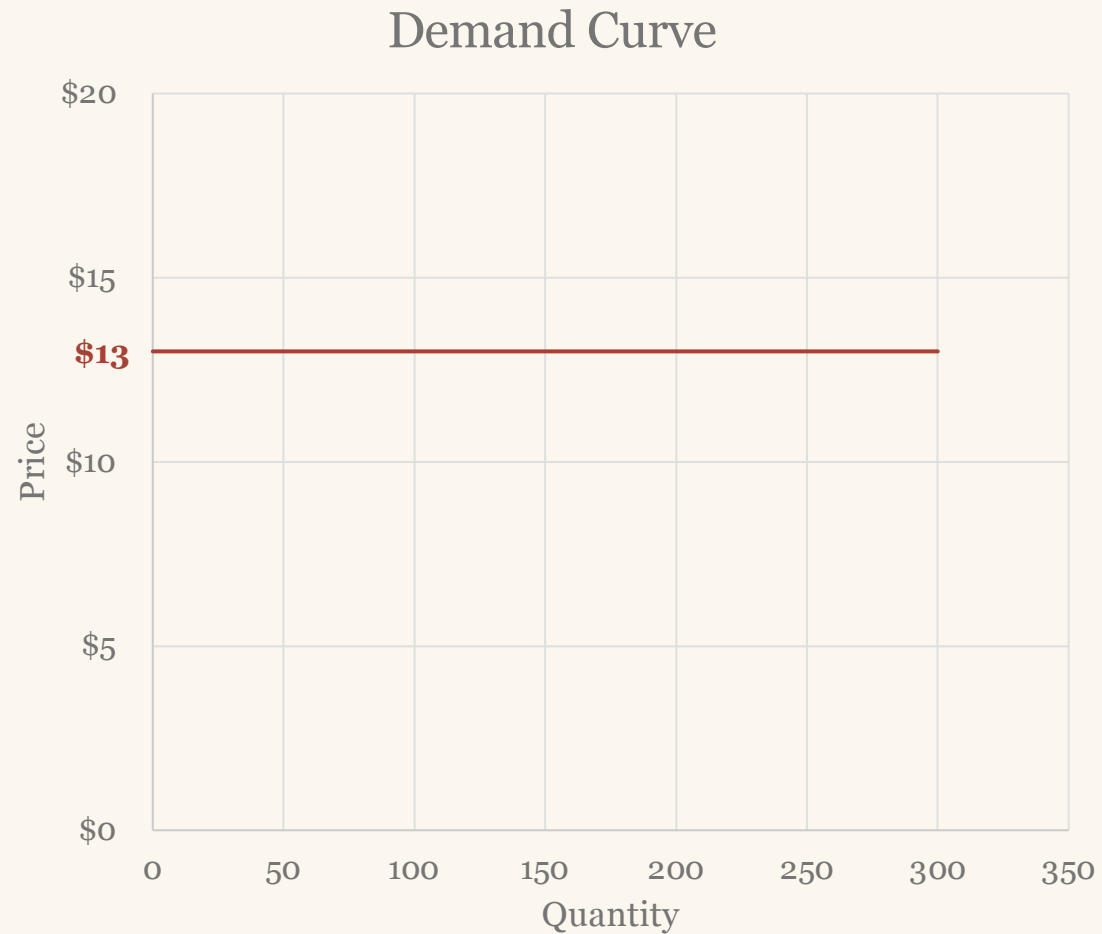
The key question:

How much market power does one seller have?

Perfect competition is the benchmark:

One seller has almost none.

Price Takers

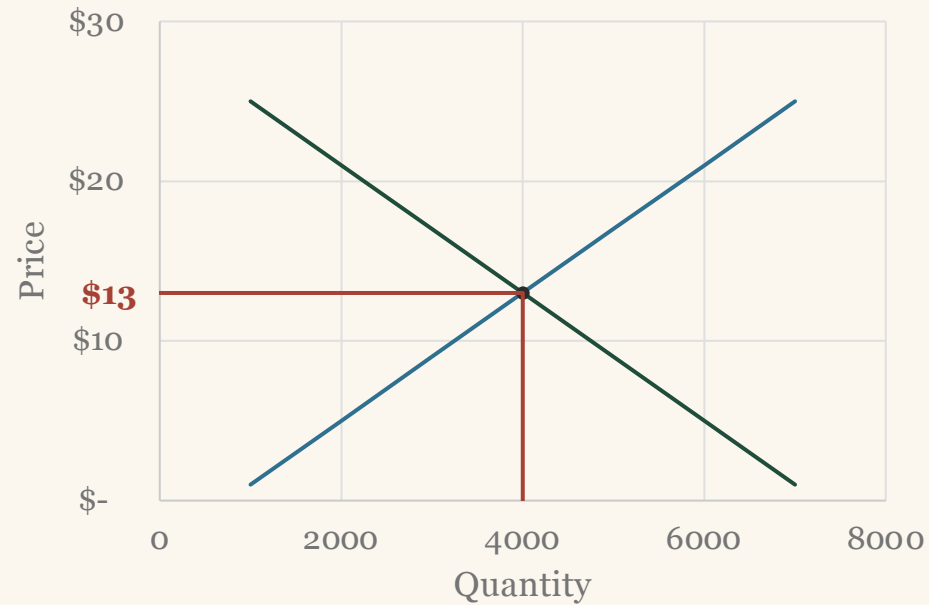


Perfectly competitive firm faces a horizontal demand curve.

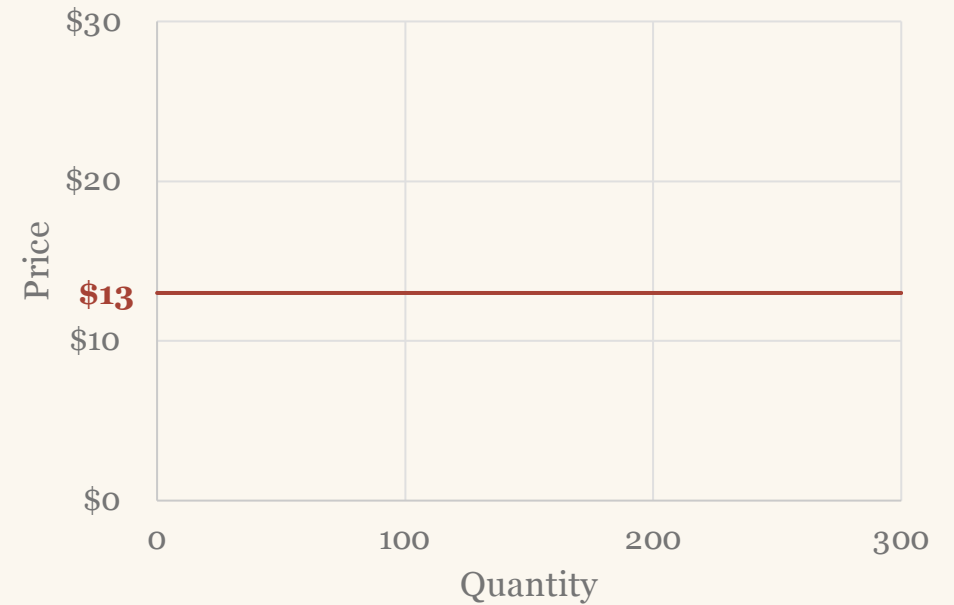
They are *price takers*.

Too small compared to the market to affect the price.

Market Demand vs Individual Demand

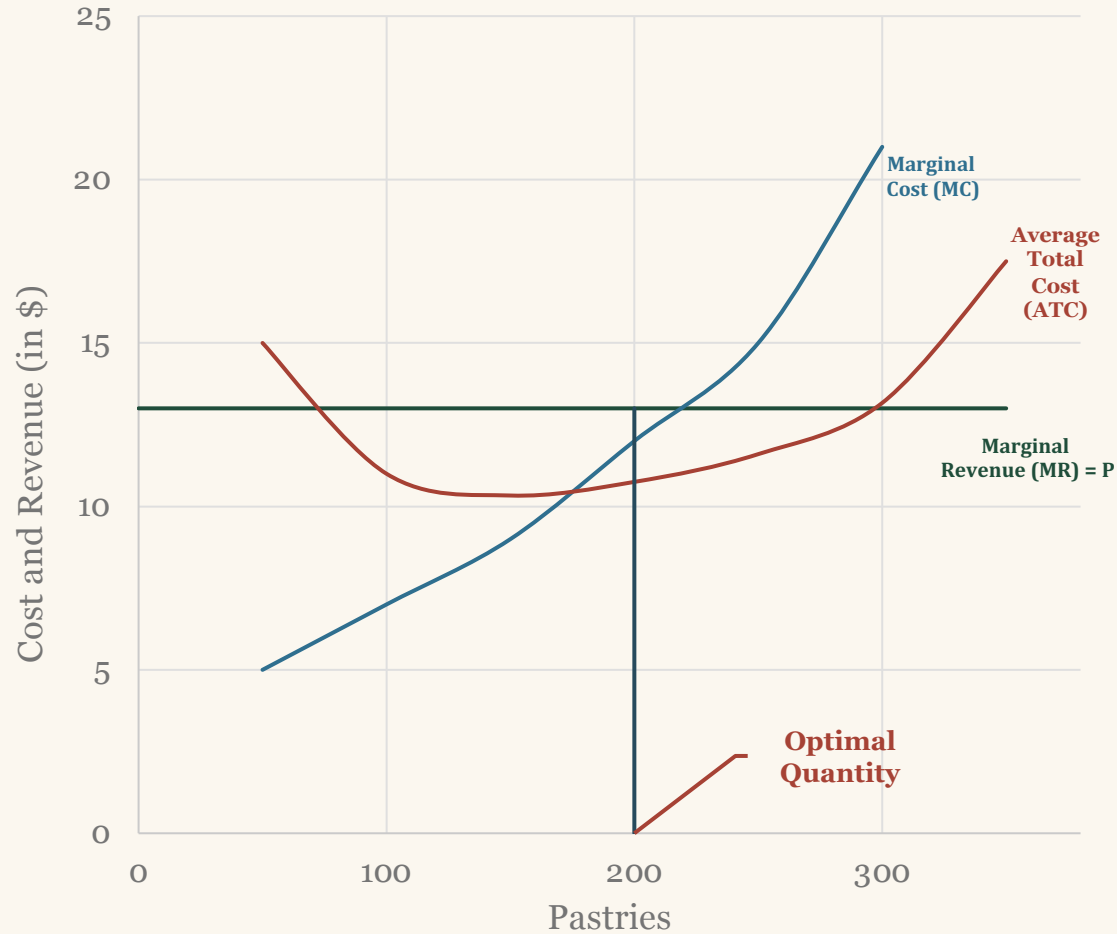


Market Demand



Individual Demand

The Profit Box



$$\text{Profit of a firm} = P \times Q - TC$$

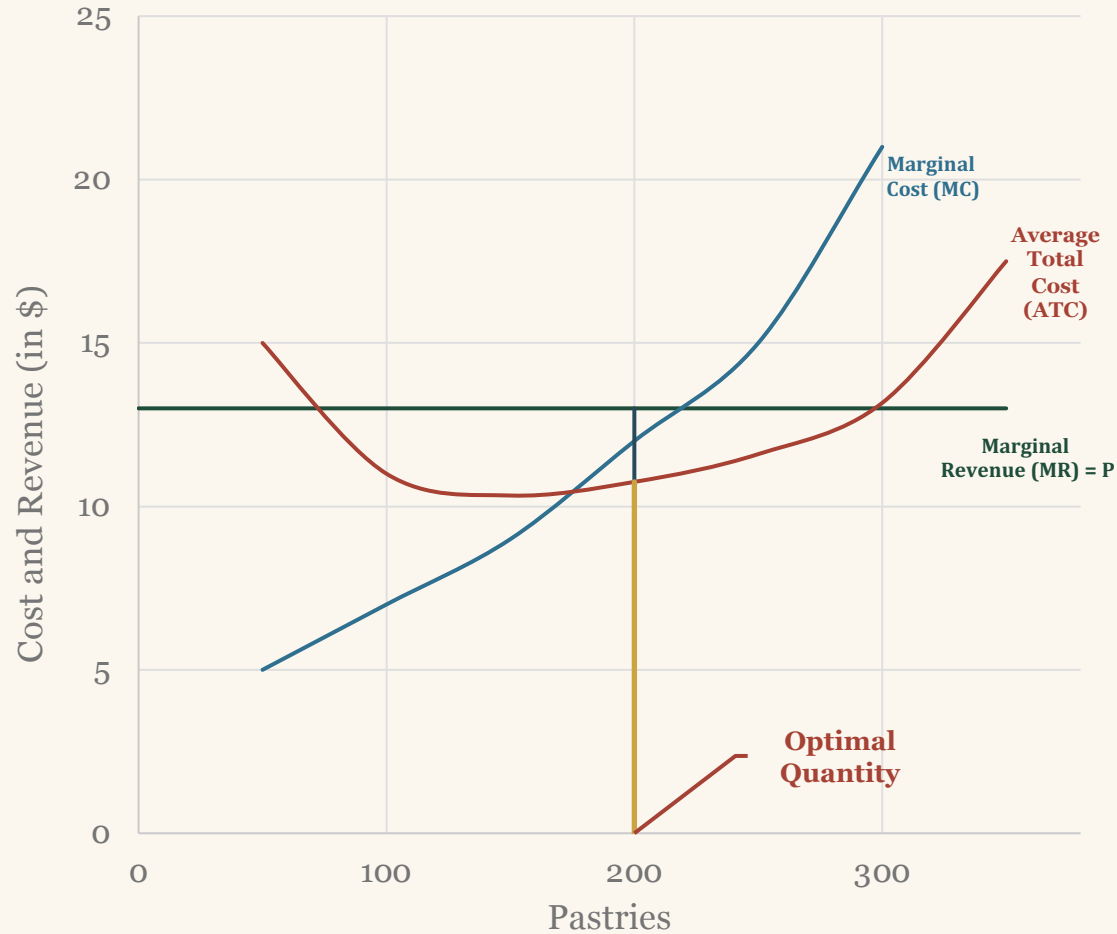
Dividing both sides by Q:

$$\frac{\text{Profit}}{Q} = P - \frac{TC}{Q} = P - ATC$$

And hence, we have:

$$\text{Profit} = (P - ATC) \times Q$$

The Profit Box



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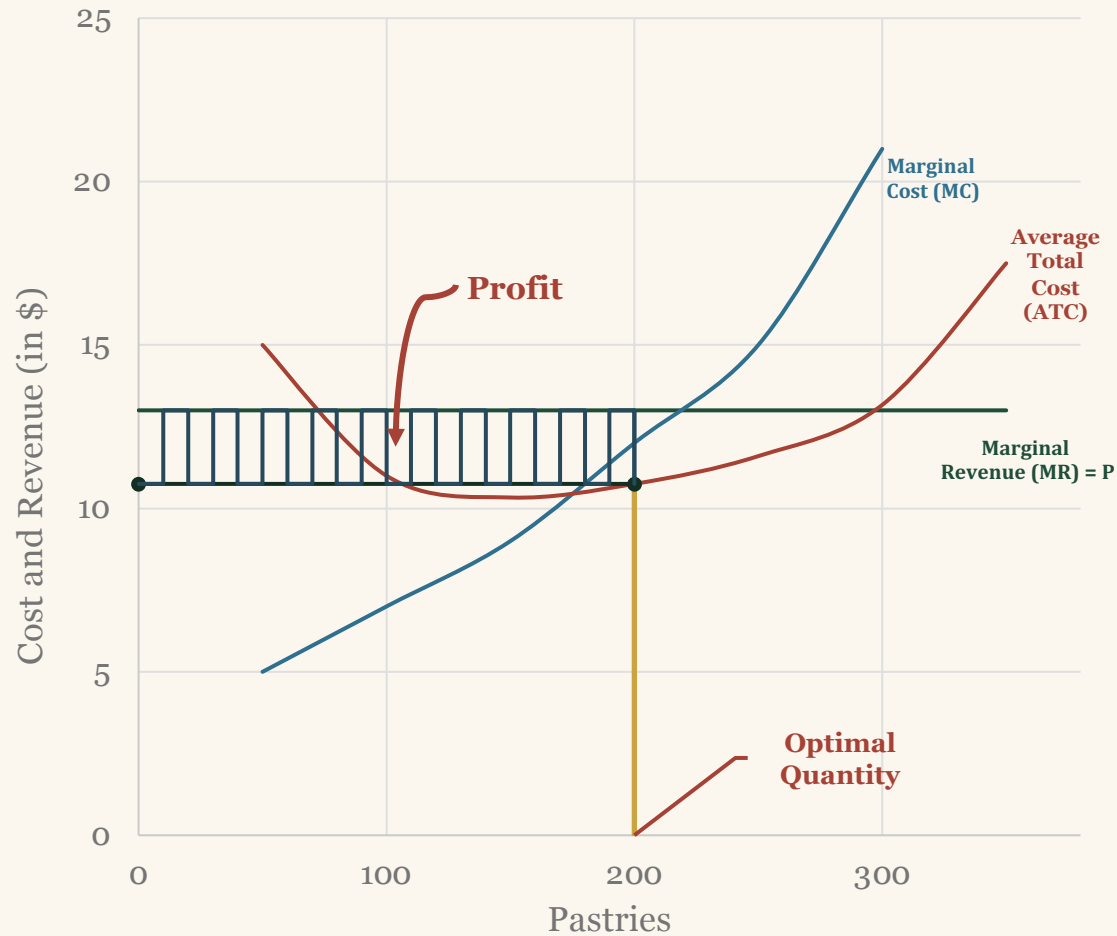
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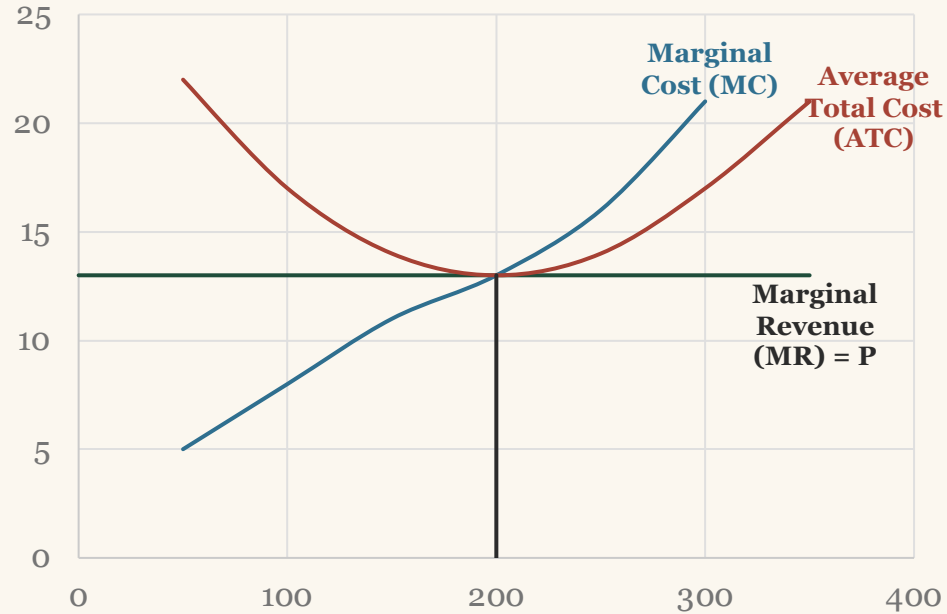
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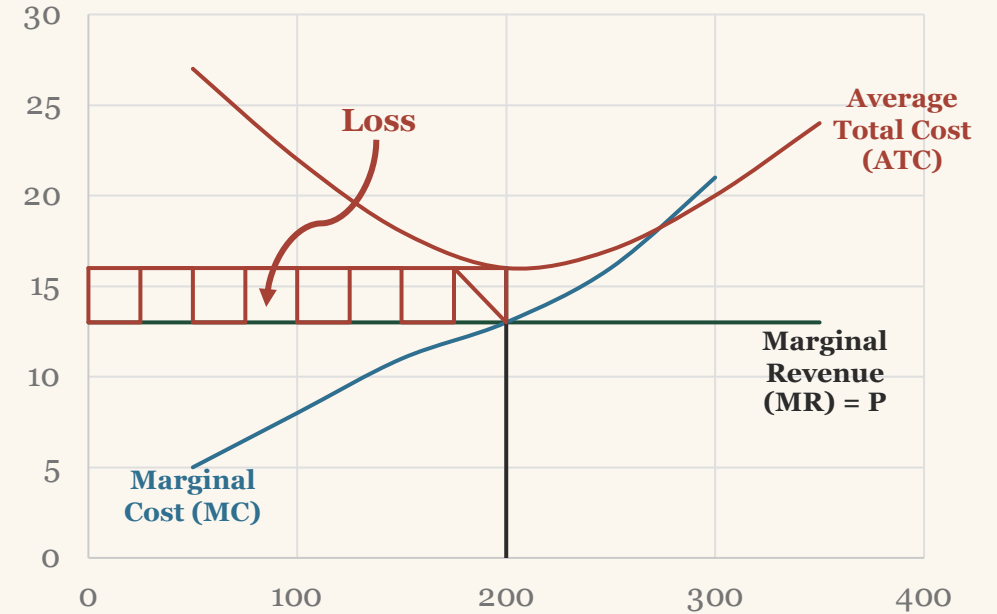
And hence, we have:

$$\text{Profit} = (P - ATC) \times Q$$

Two Different Cases



The Firm is **Breaking Even**,
 $MR = MC = ATC$
Profit = 0



The Firm is **Making a Loss**
 $MR = MC < ATC$
Profit < 0

Should the Stall Open Today?

A loss does not always mean shutting down.

Some costs are already paid (Fixed Costs):

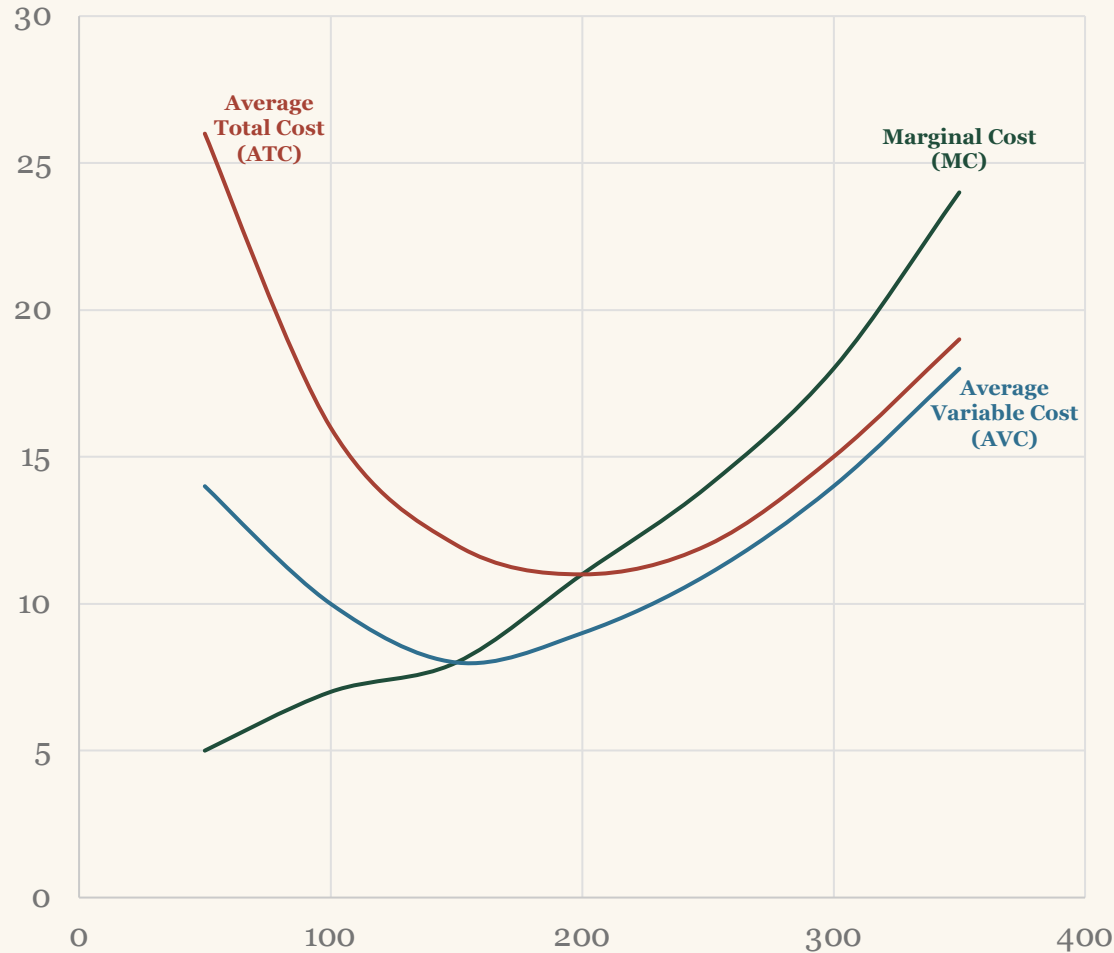
booth fee

oven rent

Permit

Can today's revenue cover today's Variable Costs?

The Shutdown Rule



In the short run:

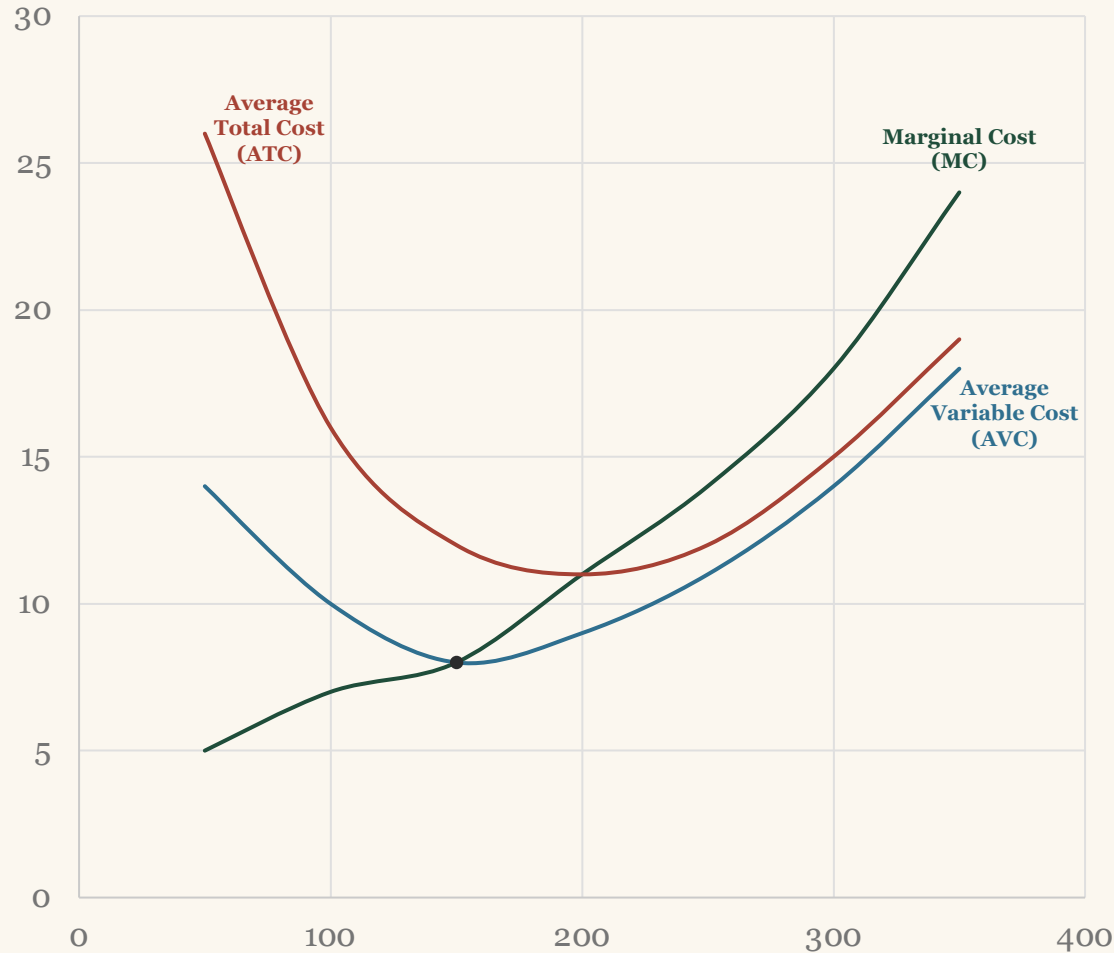
Produce if: $P \geq AVC$

Shut down if: $P < AVC$

Shutdown means producing nothing,
 $Q = 0$

Shutdown Price: $P_{min} = AVC$

The Shutdown Rule



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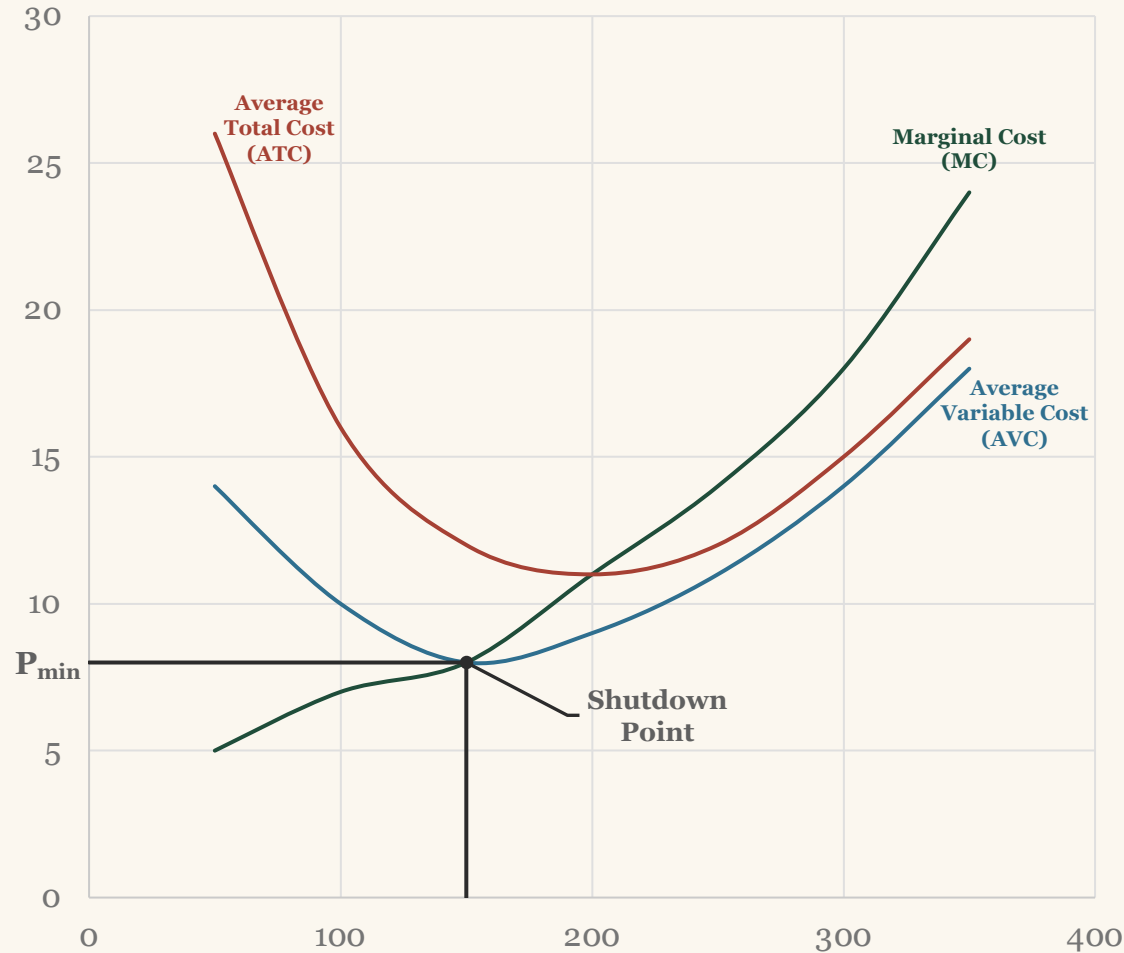
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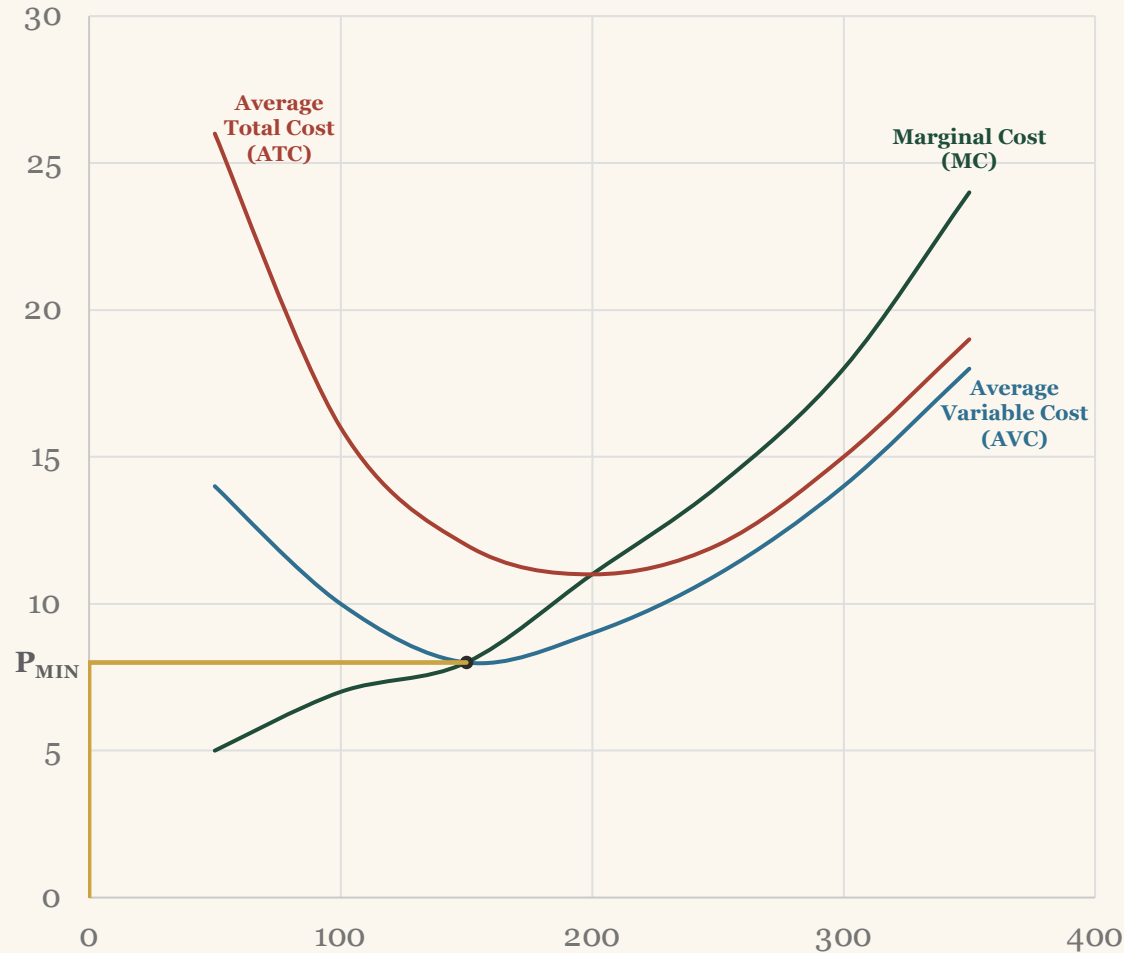
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The Stall's Supply Curve



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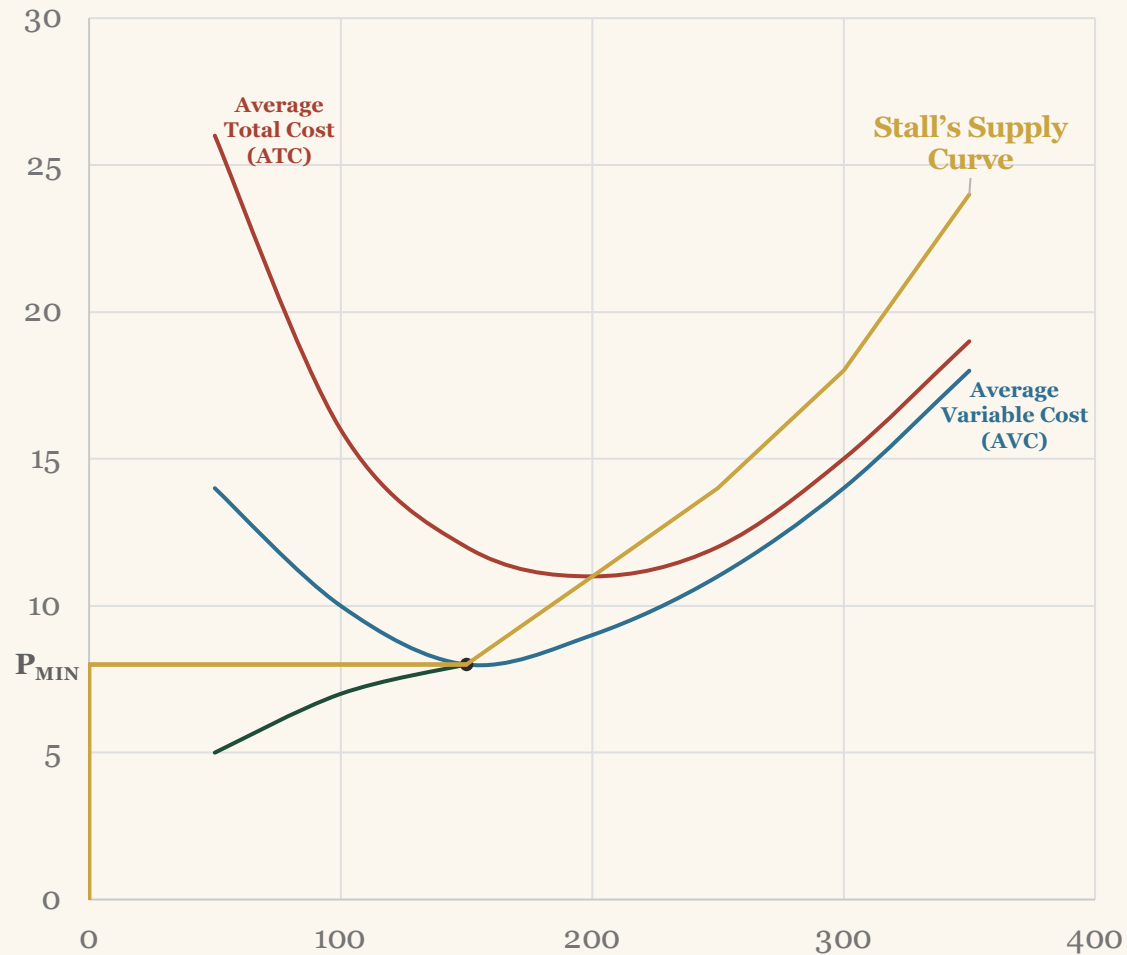
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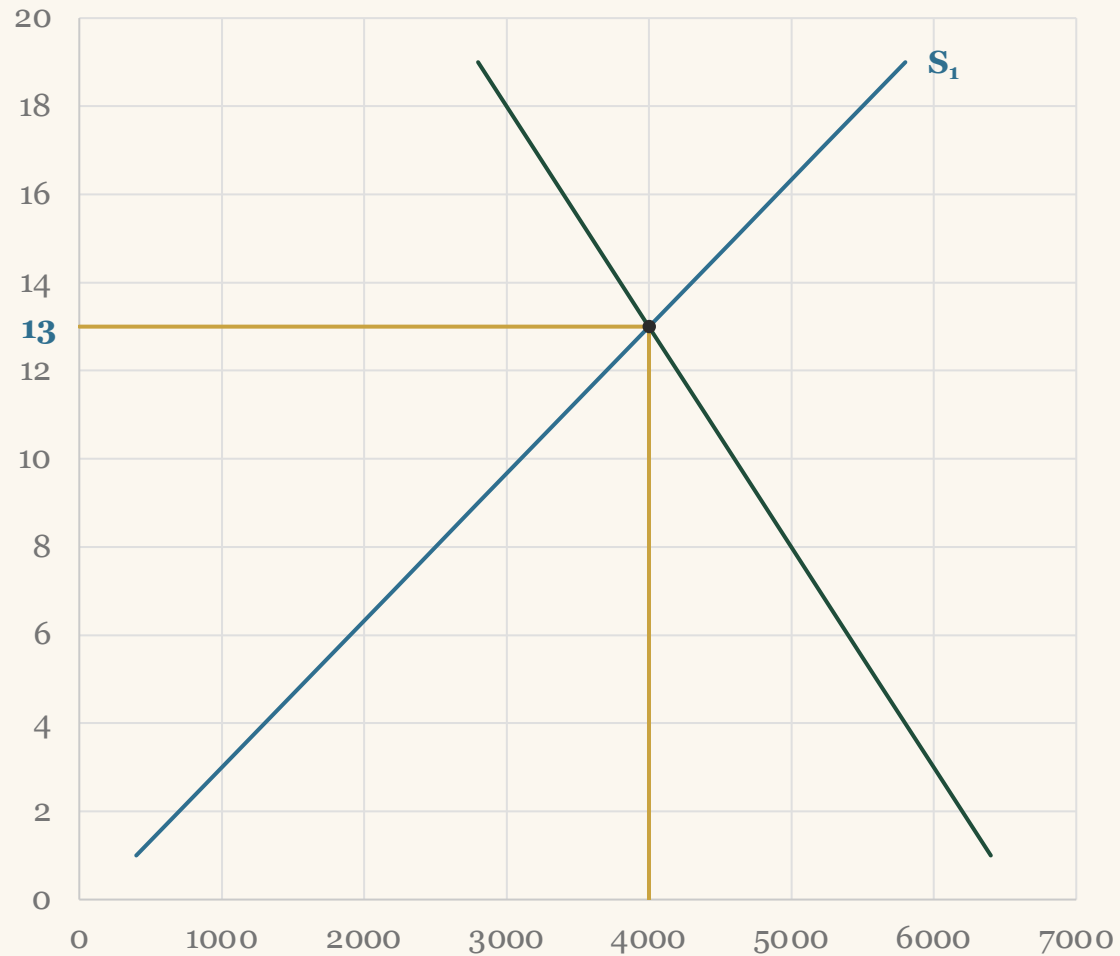
Shutdown Price: $P_{\min} = AVC$

Profit Invites Entry

Suppose Stalls earn economic profit.

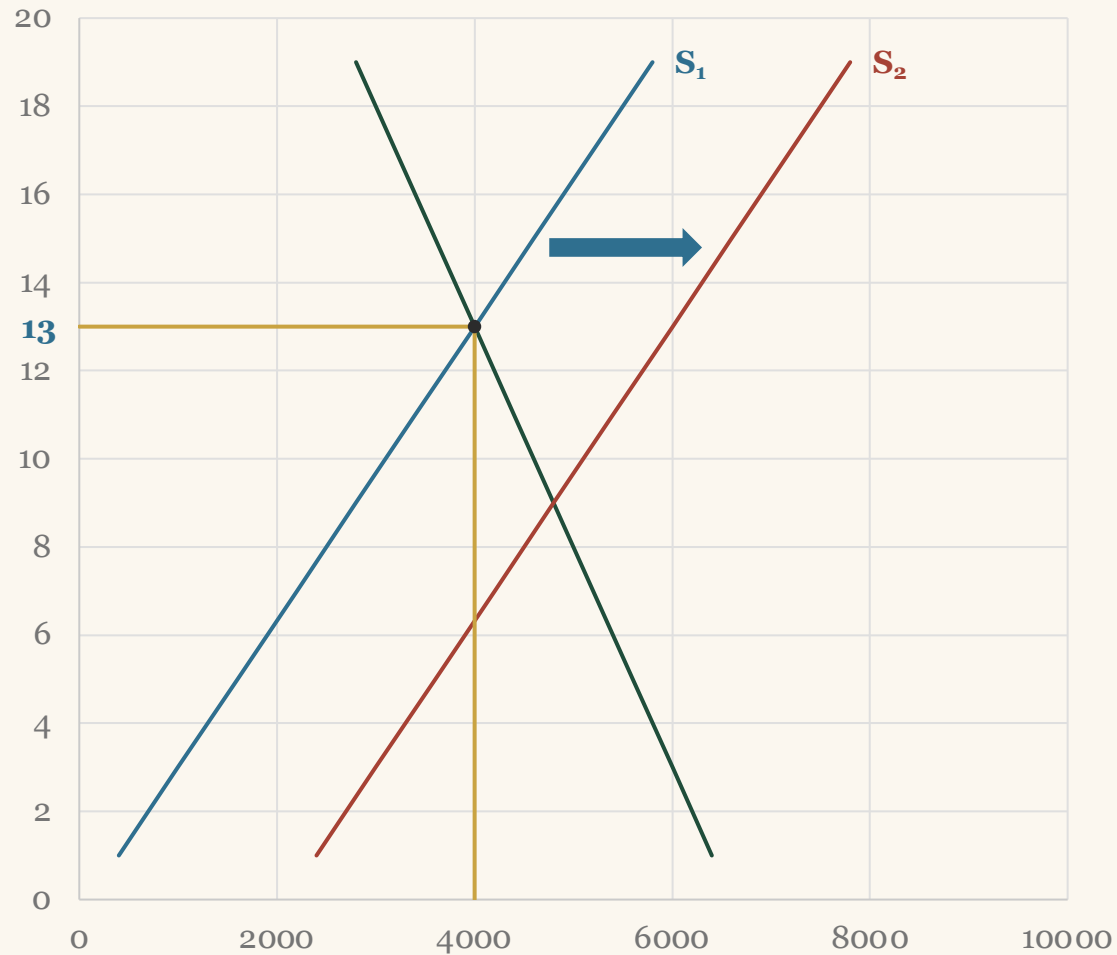
Profit sends an invitation:
“Come sell pastries here.”

Profit Invites Entry



New sellers enter

Profit Invites Entry

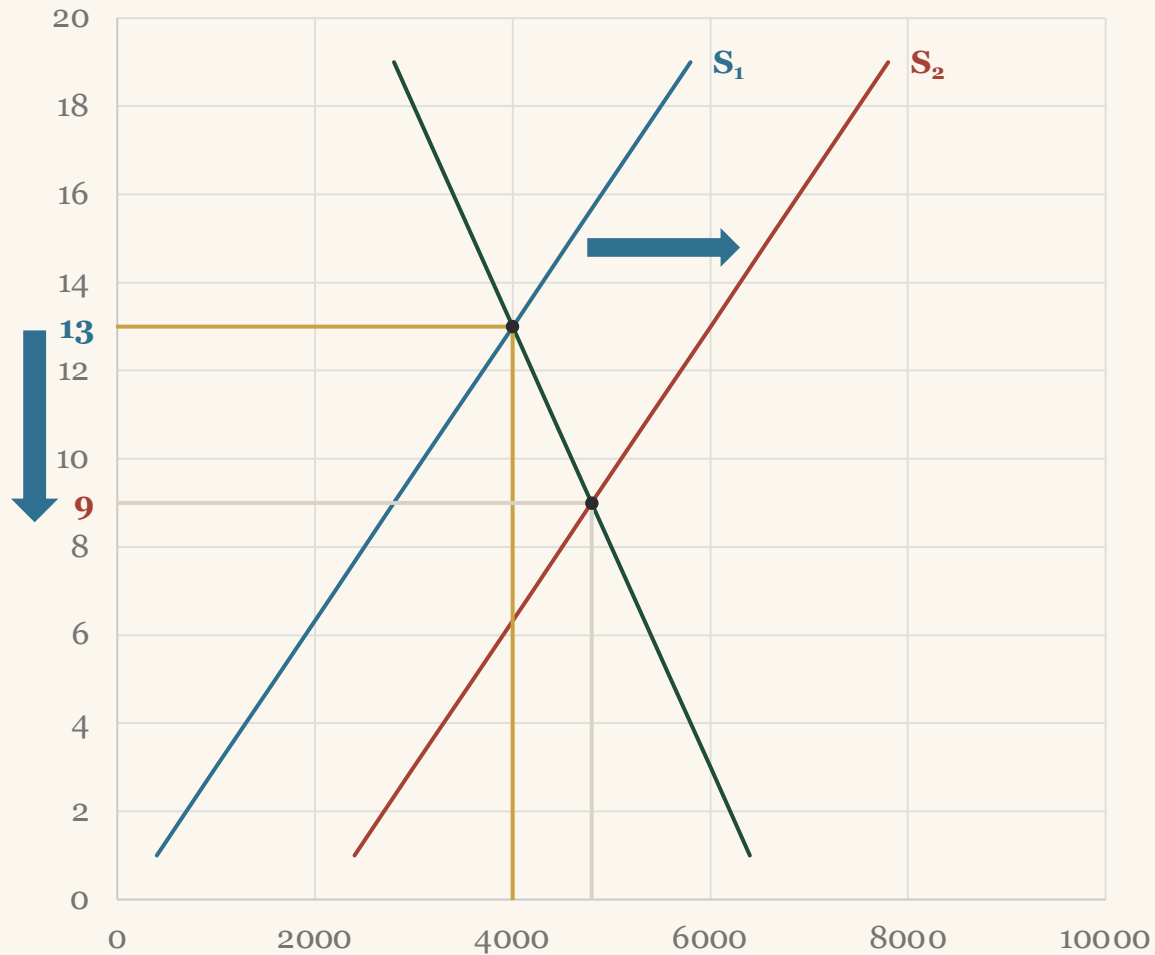


New sellers enter



Market supply increases

Profit Invites Entry



New sellers enter

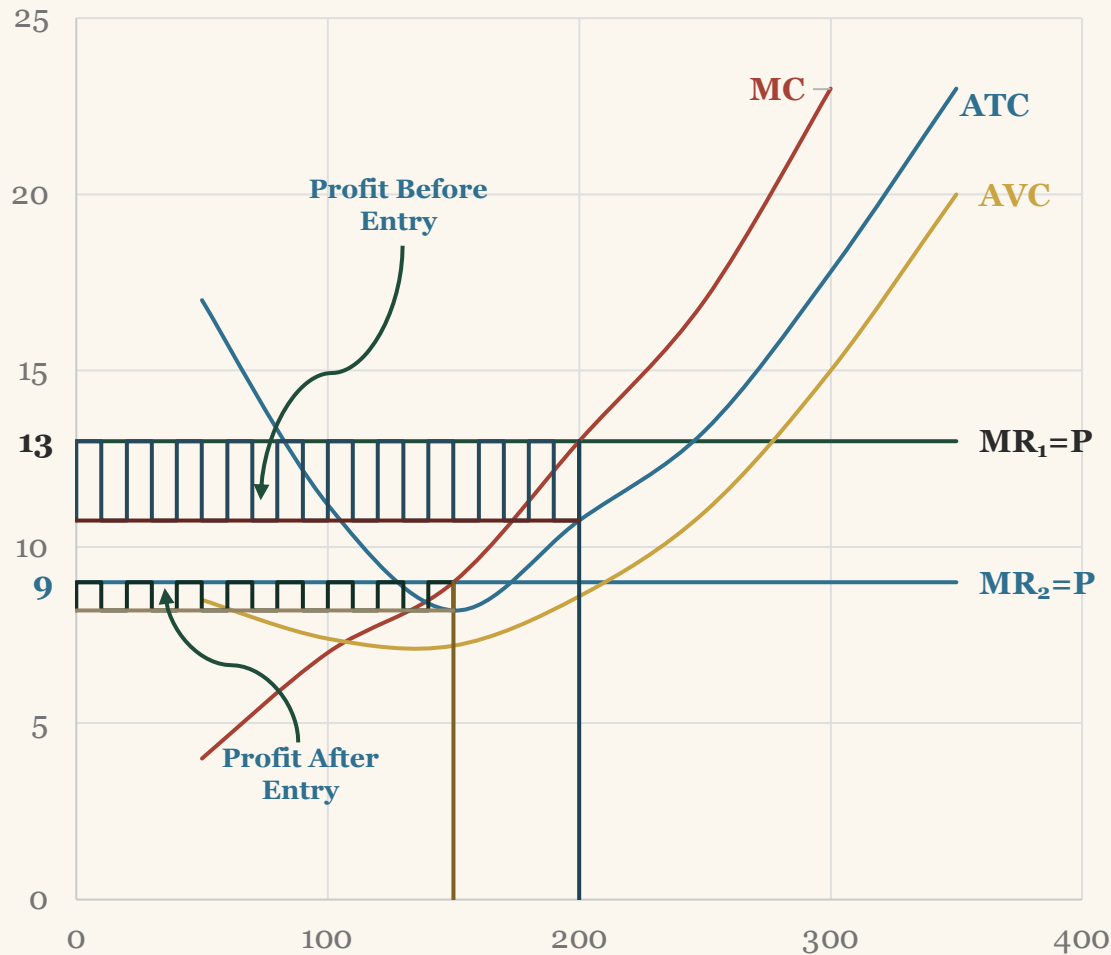


Market supply increases



Price falls

Profit Invites Entry



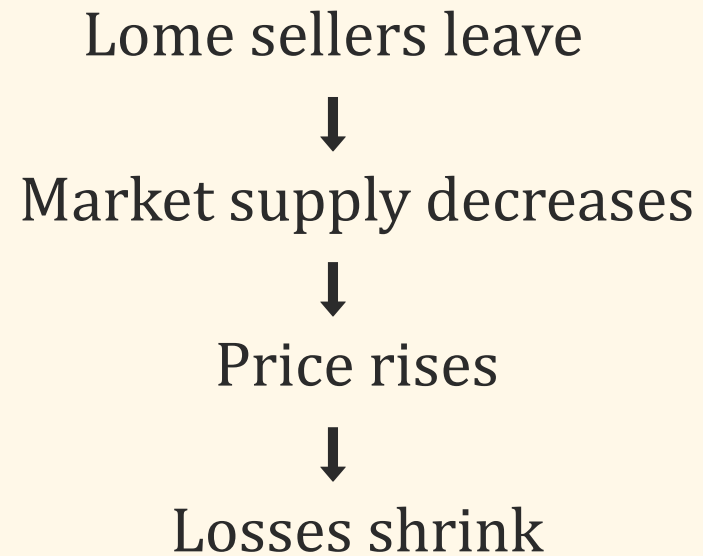
New sellers enter
↓
Market supply increases
↓
Price falls
↓
Profit shrinks

Loss Pushes Exit

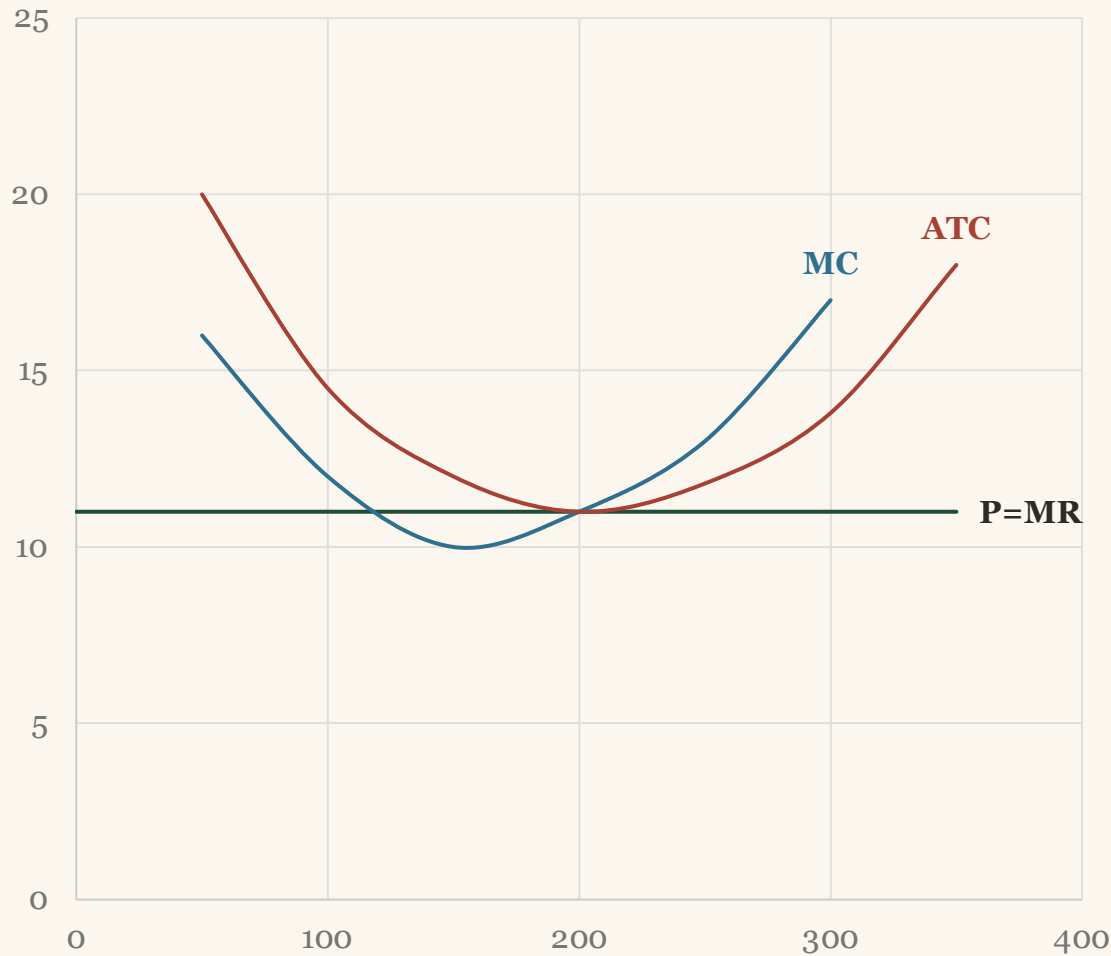
Suppose stalls keep making losses.

Loss sends a warning:
“Maybe this market is not worth staying in.”

Loss Pushes Exit



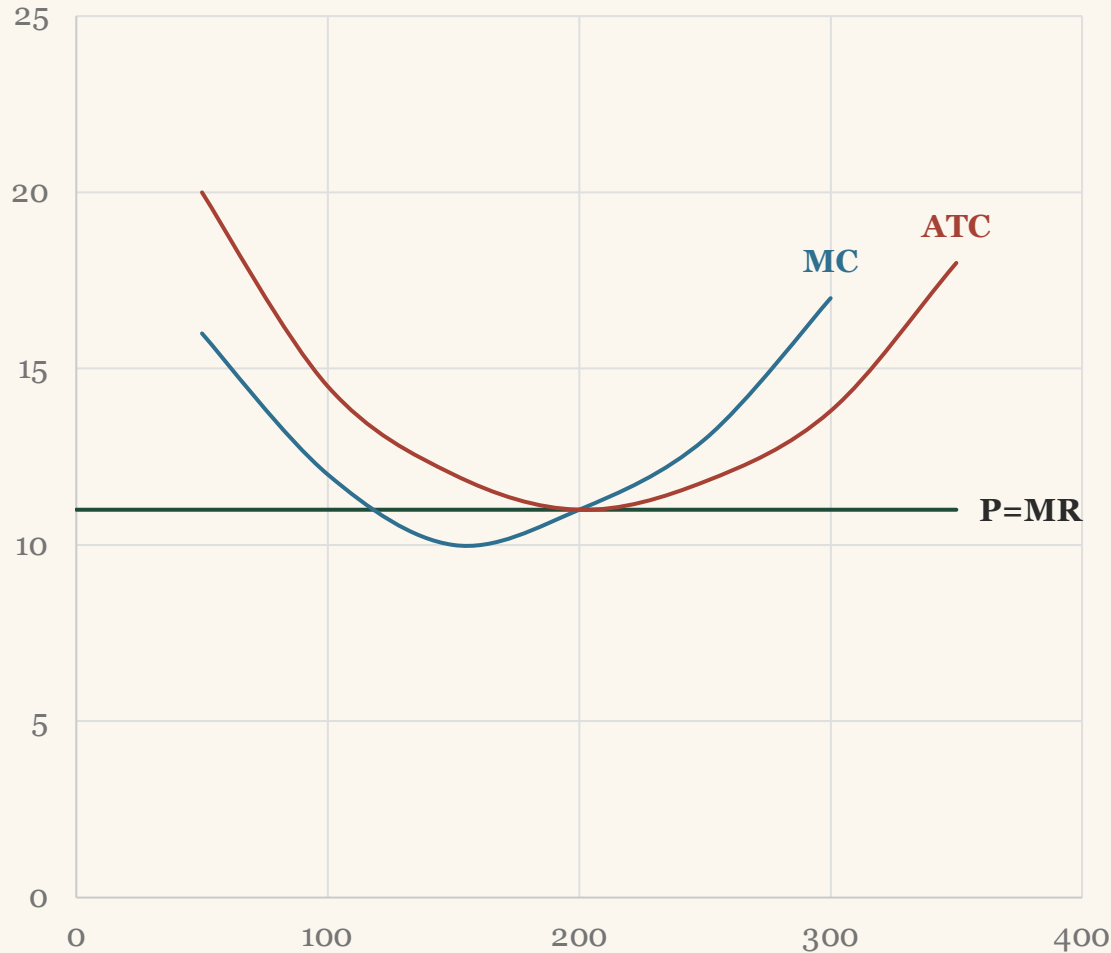
The Long Run Farmers' Market



Entry and exit stop when:

$$P = MC = \text{minimum ATC}$$

The Long Run Farmers' Market



At this point:

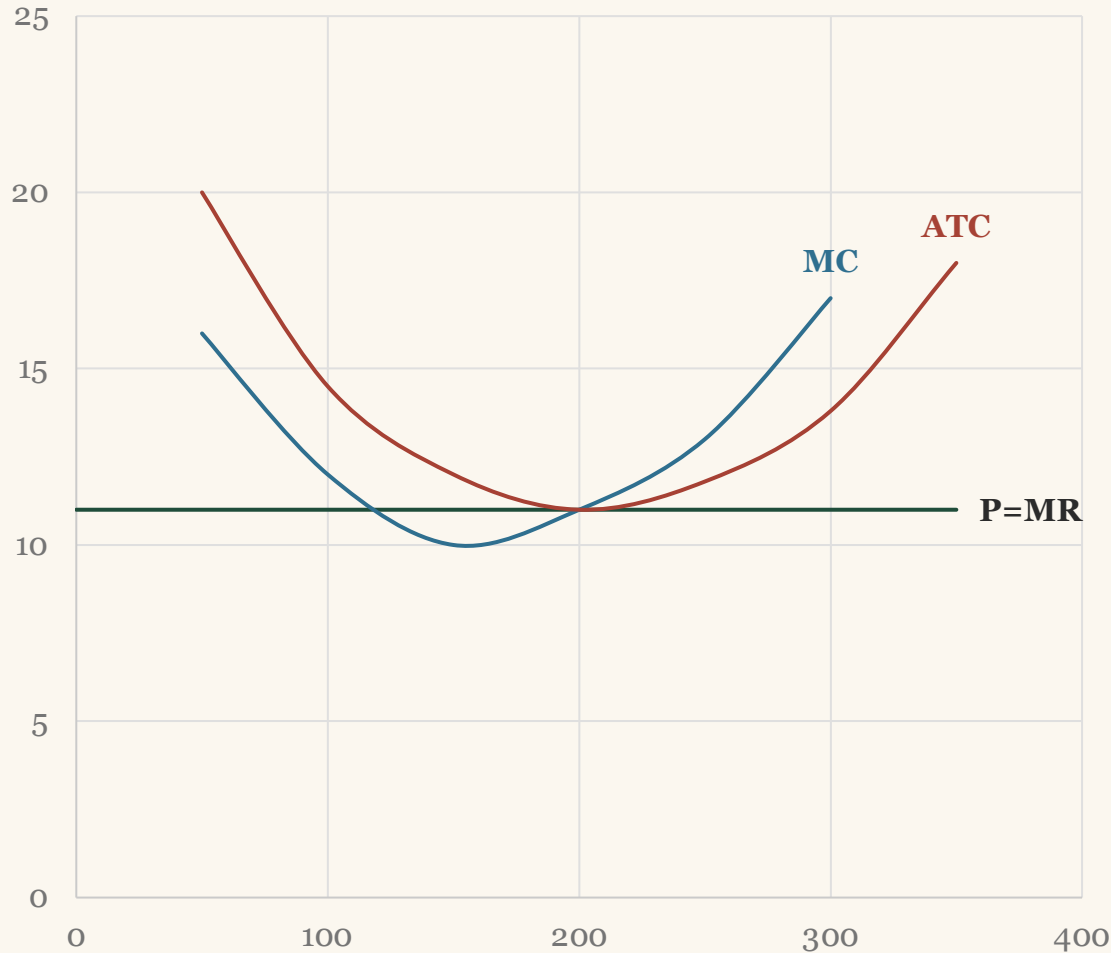
firms break even

no one wants to enter

no one wants to leave

goods are produced at lowest
average cost

The Long Run Farmers' Market



Entry and exit stop when:
 $P = MC = \text{minimum ATC}$

This is the long-run **competitive equilibrium**.